Group 9 Proposed Solution and Subsytem Breakdown

Proposed Solution:

A low-power, deployable autonomous buoy capable of continuously measuring and transmitting ice floe dynamics and environmental conditions from the Antarctic Marginal Ice Zone.

Subsystem Breakdown:

Maarij will be doing the sensing subsystem which is responsible for the acquisition of physical data from the Antarctic Marginal Ice Zone via an array of sensors interfaced with a central microcontroller.

Ebrahim will be doing the enclosure subsystem which includes selection of materials to be used as well as geometric design of the enclosure such that it survives on both ice and in the ocean.

Boitseko will be doing the communications subsystem which involves the multi-criteria-based selection of radio technologies within a mesh network.

Josh will be doing the power subsystem which will involve battery selection and battery thermal management